

TABLE OF CONTENTS

E-211 MEDIUM VOLTAGE MOTORS

	Page No.
1. SCOPE OF WORK.....	E-211-2
2. CODES AND STANDARDS	E-211-2
2.1 National.....	E-211-2
2.2 International Electrotechnical Commission.....	E-211-2
2.3 American National Standards Institute	E-211-2
2.4 Institute of Electrical and Electronics Engineers	E-211-2
2.5 National Fire Protection Association	E-211-2
2.6 National Electrical Manufacturers Association.....	E-211-3
3. OPERATING AND DESIGN CONDITIONS	E-211-3
4. DETAILED REQUIREMENTS	E-211-3
4.1 General	E-211-3
4.2 Ratings	E-211-4
4.3 Insulation.....	E-211-4
4.4 Torque Requirements	E-211-4
4.5 Enclosures	E-211-5
4.6 Bearings	E-211-6
4.7 Terminal Boxes	E-211-8
4.8 Winding Temperature Detectors	E-211-9
4.9 Direction of Rotation and Terminal Markings	E-211-9
4.10 Nameplates.....	E-211-9
4.11 Service and Life.....	E-211-10
4.12 Margin for Relaying	E-211-10
4.13 Motor Surge Protection.....	E-211-10
5. PERFORMANCE TESTING AND GUARANTEES.....	E-211-10
5.1 Type Tests.....	E-211-10
5.2 Routine Tests	E-211-10
6. QUALITY ASSURANCE/QUALITY CONTROL	E-211-11

E-211 MEDIUM VOLTAGE MOTORS

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of all Medium Voltage Motors in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards, and the following Specific Standards:

2.1 National

PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation – Indonesia).

SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

IEC 60034 : Rotating electrical machines

IEC 60445 : Basic and safety principles for man-machine interface, marking and identification – Identification of equipment terminals and of terminations of certain designated conductors, including general rules for an alphanumeric system.

2.3 American National Standards Institute

ANSI C 50.41 : Standard for Poly-phase Induction Motors for Power Generating Stations.

2.4 Institute of Electrical and Electronics Engineers

IEEE 112 : Test Procedure for Poly-phase Induction Motors and Generators

2.5 National Fire Protection Association

NFPA 70 : National Electrical Code

2.6 National Electrical Manufacturers Association

NEMA MG-1-22 : Motors and Generators

3. OPERATING AND DESIGN CONDITIONS

Motors shall be suitable for operation in a gas fired combined cycle Power Plant located in a tropical climate along the sea coast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for the meteorological, seismic and environmental conditions at the Site.

Motors shall deliver their rated kW output, without damage, with $\pm 10\%$ variation of rated voltage at the terminals at rated frequency. Motors shall deliver their rated kW output, without damage, with $\pm 5\%$ variation in frequency.

Motors shall deliver their rated kW output, without damage, when the sum of the simultaneous variation of voltage and frequency does not exceed $\pm 10\%$.

4. DETAILED REQUIREMENTS

4.1 General

Motors shall be heavy duty, having highest possible efficiency and power factor compatible with good motor design.

Motors shall be supplied with anti-condensation heaters suitable for a 230 V AC single phase 50 Hz supply. Heaters shall be automatically de-energized when the motor is switched on and automatically energized when the motor is not operating.

Each motor shall be provided with grounding connections. The grounding connections shall not be located on a removable part of the motor.

The front end of the shaft (opposite the driving end) shall be centered and shall be accessible for taking speed readings with a portable tachometer.

All screws, bolts, nuts, springs, studs and other miscellaneous fittings shall be of corrosion-resistant material.

Motors shall be designed, built and adequately braced for across-the-line starting with 80 to 110% of full rated voltage applied at the motor terminals.

4.2 Ratings

Motors shall have 1.0 service factor. The nameplate ratings shall not be less than 115% of the requirements of the driven equipment when operating under design requirements and not less than 100% of the power required to operate the driven equipment at its maximum capability. This relation shall apply for all operating speeds and conditions.

Motors shall be rated for continuous operation in a 40°C ambient temperature. The rated temperature rise shall be correspondingly reduced in accordance with IEC 60034.

4.3 Insulation

Motor insulation shall be Class F with Class B insulation design temperature rise. The motor insulation shall be treated for fungus and abrasion resistance.

4.4 Torque Requirements

Motors shall be designed to have sufficient torque for starting and accelerating the maximum load when 80% of motor rated voltage is applied to the motor terminals, without exceeding the thermal limits of the motor. The permissive locked rotor time shall be greater than the accelerating time with the motor loaded and with motor terminal voltage at 80% rated voltage. The reduced voltage condition shall be considered to exist over the full accelerating period.

The accelerating torque at any speed and 80% rated voltage shall not be less than 10% of motor rated full load torque. The motor starting torque at 100% rated voltage shall not be less than rated torque at rated speed. The Contractor shall submit the motor starting method required, the motor combined thermal limit and accelerating current - time curves (80% and 100% voltage) as shown in (Figure 1 attached) and speed vs torque curves with superimposed load torque curves (80% and 100% voltage) as shown in (Figure 2 attached) with the Bid Proposal.

Breakdown torque shall be adequate to maintain motor operation during a voltage dip to 75% of rated voltage for period of 10 seconds at rated load and frequency.

Squirrel-cage induction motors, operating with rated terminal voltage and rated frequency shall have torque values in accordance with the applicable IEC standards and/or requirements determined by the nature of driven equipment with minimum starting voltage and voltage dips, while the motor is running.

Motor shall have the lowest locked rotor current values consistent with high efficiency and good performance designed motors in their torque-current class.

However locked rotor current, in percent of full load current, shall not exceed the following values unless agreed to by the Owner.

- (1) 500% for motors above 3500 kW.
- (2) 600% for motors 600 kW up to 3500 kW.
- (3) 650% for motors smaller than 600 kW.

4.5 Enclosures

Medium Voltage Motors located indoors shall be drip proof with IP 23 degree of protection.

Medium Voltage Motors located outdoors shall be weather protected with an IPW 54 degree of protection meeting the requirement for NEMA WPII enclosures. Outdoor motors shall be provided with air filters and seals at the shafts, terminal box, and around motor terminal leads, to prevent ingress water, when subject to stream of water from a nozzle with pressure of 2.8 kg/cm². Motors which may be exposed to direct sun-light, shall be designed for that service. Sun shields shall not be used.

Motor located in hazardous areas as defined in NFPA 70 shall be explosion proof or dust-ignition proof as required.

Motors shall be provided with suitable means of drainage to prevent accumulation of water due to condensation. Explosion proof and dust-ignition proof motors shall be provided with an approved means of drainage.

Housings shall be cast iron or steel plate. Aluminum shall not be used.

The Bidder's recommendation as to which type of motor will be provided for a particular application, shall be subject to the Owner review.

Motors shall be provided with a dial type thermometer with a telltale maximum temperature pointer and adjustable alarm contacts to measure the highest temperature of the internal cooling air in contact with the windings. The alarm

contacts shall be suitable for remote indication. Alarm contacts shall be two (2) electrically independent contacts.

4.6 Bearings

Unless otherwise approved, bearing lubrication shall be of the same type as provided for the driven equipment and where forced lubrication is used, the system shall be common to the motor and driven equipment.

Motors with sleeve bearings shall be provided with the followings:

- (1) Shaft and bearing housings shall be constructed with oil deflectors and seals so that the lubricant will not leak along the shaft or be thrown into the windings.
- (2) Sleeve bearings shall be ring lubricated and furnished with oil ring sight opening and with suitable oil gauges or covered overflow cups. Covers for oil wells and overflows shall be attached with chains or be hinged. Motors that have a forced oil lubrication system shall have oil rings that provide for a short time operation without the lubrication system in operation. Specific recommendations shall be given by the motor manufacturer regarding lubricants and lubrication.
- (3) When oil-level sight-gauges are furnished they shall be marked to show proper oil level for standstill and for running conditions when the difference in levels is appreciable.
- (4) Bearing temperature thermocouple with connection heads shall be furnished. The elements shall be installed as close as possible to sleeve bearing surfaces and shall be located preferably in the bottom half of the bearing housing. Refer to Clause 4.7 for termination requirements.
- (5) The motor manufacturer shall:
 - a. Indicate on the motor outline drawing the minimum rotor end- play.
 - b. Mark rotor end-play limits on motor shaft.
 - c. Include means for indicating the shaft magnetic center position.

- (6) Motors with sleeve bearings and end shields, exclusive of explosion proof motors, shall be provided with two access holes in at least one end shield, located near the bottom and opposite the air gap, to permit insertion of an air gap or feeler gauge between the rotor and stator.

Motors with ball or roller type anti-friction bearings shall be provided with the followings:

- (1) Bearing housing shall be large enough to hold sufficient lubricant to minimize the need for frequent lubrication. Motors shall be shipped with housings properly packed with manufacturer's recommended lubricant and attached instruction cards for adjustment required prior to operation. The bearings shall be equipped with Zerk grease fittings (or approved equal) and automatic grease reliefs.
- (2) Anti-friction bearings shall be provided with suitable grease and relief fittings so that lubricant can be forced through the bearing housings and also for positive purging of the old grease, without disassembling of motor parts.
- (3) Anti-friction bearings for direct coupled motors shall have minimum L-10 rating life of 150,000 hours in accordance with AFBMA. (The rating life is defined as the total number of operating hours, at specified bearing loading and speed conditions, that each member of 90% of a group of apparently identical bearings will complete before failure).
- (4) Motors with anti-friction bearings shall be provided with vibration detectors where specified or required. Refer to Technical Specification I-001 for vibration detector requirements and Clause 4.7 herein for termination requirements.

To prevent shaft circulating currents one or both bearings shall be insulated from motor frame. Manufacturer's motor drawings shall clearly indicate when piping and conduit to bearings must be insulated to protect against the occurrence of shaft currents.

Horizontal Motors

Horizontal motors, equipped with sleeve bearings and end shields or self-contained bearing brackets shall have horizontal split end shield or bracket bearing housings and bearing on the coupling end, except that belted motors and explosionproof motors shall have solid end shields on both ends. A removable bearing supporting plate attached to the end shield, providing an opening through which rotor and coupling may be withdrawn, is acceptable.

Horizontal motors shall have bearing housings large enough to hold sufficient lubricant to minimize the need for frequent lubrication and shall have convenient facilities for filling and draining.

Vertical Motors

Vertical motors shall be provided with double pivoted Michell or Kingsbury type thrust bearing (or approved equal) mounted in the motor frame and designed to support the combined weight of the rotor, the shaft mounted driven equipment and the thrust from the driven equipment in the same or the opposite direction.

All bearings shall be designed for the maximum overspeed duty requirements in the reverse direction imposed by reverse flow through the driven equipment.

The guide bearings shall preferably be of the sleeve type. They shall be equipped with easy to see oil level indicators and easily accessible oil drainage facilities.

4.7 Terminal Boxes

Terminal box for motor leads shall be waterproof, corrosion resistant stainless steel with non-magnetic gland plates, supported phase leads and provided with pressure relief means capable of withstanding minimum 40 kA, rms short circuit current. Gland plates shall be bolted and gasketed removable plates suitable for drilling of conduit entrance holes in the field.

Motor lead terminal boxes shall be located on the right hand side of motor, when viewed from the drive end, unless equipment location dictates otherwise as reviewed by the Owner. Terminal boxes shall be suitable for mounting in all 90 degree positions without alteration of terminal box or bolting facilities so that conduit may enter from above, below or from either side. Terminal boxes shall be located so that conduit will clear the motor base by at least 80 mm, without bending the conduit, when conduit enters from below.

Motor terminal leads shall be provided with connection assembly with lugs, for termination of motor feed cables. The contractor shall furnish lugs for the motor feed cables.

Leads from motor space heaters, bearing thermocouple and/or vibration detectors and stator temperature detectors shall be brought out into their own separate terminal boxes. The detector leads shall be terminated at suitable terminal blocks, mounted inside the terminal boxes. Accessory leads or terminal blocks shall be permanently marked with proper designations.

All terminal boxes shall be positively grounded to the motor housing and also they shall be provided with grounding terminals inside the terminal box housings for cable shield groundings.

4.8 Winding Temperature Detectors

Motors shall be provided with stator winding temperature detectors, which shall consist of six (6) resistance temperature detectors (RTDs), 100 ohm two (2) RTDs per phase, embedded in the stator windings.

4.9 Direction of Rotation and Terminal Markings

The connections to the motors shall be such that when the supply leads, L1, L2 and L3 are connected to motor terminals, U,V,W respectively, the motor shall rotate in a clockwise direction when viewed from the driving end. Where such motors require anti- clockwise rotation, the supply leads L1, L2, and L3 shall be connected to terminals U, W, and V respectively, in accordance with IEC 60445.

Motor leads shall be permanently marked with motor phase identification alphanumeric designations.

4.10 Nameplates

A nameplate shall be provided for each motor. The nameplate shall list all motor data required by IEC 60034 and the following:

- (1) Service factor
- (2) Shaft rotation corresponding to a supply voltage sequence of U - V - W
- (3) Lubricant type and lubrication instructions
- (4) Terminal arrangements
- (5) Maximum permissible number of starts and the required cooling period

when the motor is started under cold and hot motor conditions.

4.11 Service and Life

Motors will be used to operate their specified loads in a power generating station. They will be operated continuously or started intermittently. The type of operation will vary over the life of the plant, however, in general the frequency of starting, will increase as the plant gets older. Motors shall be designed for thirty (30) year service life. The Bidder shall indicate in his Bid Proposal number of allowable hot and cold re-starts per day for his motors.

4.12 Margin for Relaying

Each medium voltage motor will be protected against overcurrent/overload and for undervoltage/loss of power supply. Each medium voltage motor shall have enough margin between their thermal limit curve and the 80% starting voltage current versus time curve, so that the motor can be completely protected against overcurrents with no false tripping when starting at 80% of motor rated voltage (See Figure 1 herein). The Contractor shall provide calculations to demonstrate this for motors driving high inertia loads.

4.13 Motor Surge Protection

Surge protection shall not be used on motors. The motors must be designed to accept switching surges and voltage transients without damage to the motor and insulation. Motors may be fed from 6 kV switchgears with vacuum or SF₆ circuit breakers. The motors shall have sufficient turn-to-turn insulation strength to be able to withstand the voltage impulses that occur when the motor is energized or de-energized without the use of surge protection capacitors at the motor terminals. Attachment A to this Technical Specification indicates the required impulse withstand capability for the motor insulation.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 Type Tests

- (1) Standard type tests shall be performed on one (1) motor of each type and rating, in accordance with IEC standard. The type test not need to be repeated if type test have been executed on similar motor and test certificate is available.

5.2 Routine Tests

All motors shall be subjected to the following routine tests in accordance with IEC:

- (1) Measurement of winding resistance.
- (2) No load current, power and speed at rated voltage and frequency.
- (3) Mechanical vibration.
- (4) Direction of rotation for normal phase sequence.
- (5) Insulation resistance test.
- (6) High potential test.

6. QUALITY ASSURANCE/QUALITY CONTROL

The Contractor shall fulfill the quality assurance/quality control requirements listed for individual equipment in the respective equipment Specifications and in Part 4, Section 2.7.

The quality assurance/quality control procedures shall be as specified. These shall include, but not be limited to, the followings:

- (1) Insulation and High Voltage Tests.
- (2) Cleaning and Painting.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Inspection Reports
- (2) Certified Test Reports.
- (3) All other documentation required by applicable codes and standards.

(Last page of E.211)